

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 2

Exercise 2.1 (Stability Examples).

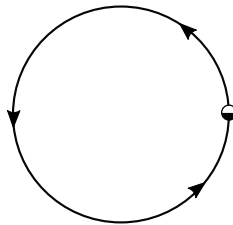
- Find a dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$ – by means of an equation or a phase portrait – which has an unstable equilibrium x_* , that satisfies $\lim_{t \rightarrow \infty} \phi_t(x_0) = x_*$ for all $x_0 \in \mathcal{X}$.
- Find another dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$, such that x^* is an asymptotically stable equilibrium, but there exists no $\varepsilon > 0$ such that $\mathcal{B}(x^*, \varepsilon)$ is positively invariant.

Solution.

- Take, for example, $\mathcal{X} = \mathbb{S}^1$, represented by some $\theta \in [0, 2\pi)$ and let the flow be induced by

$$\dot{\theta} = 1 - \cos(\theta) .$$

Then the flow is directed counterclockwise everywhere except for the equilibrium located at $\theta = 0$. Thus, every trajectory will eventually approach 0. But since small, positive initial values take almost a full rotation, their trajectories will leave any small neighbourhood of 0. This means the equilibrium at $\theta = 0$ is unstable.



- Consider the linear system

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} -\delta & -4 \\ 1/4 & -\delta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

where $0 < \delta \ll 1$. This linear system has the general solution

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} 4e^{-\delta t}(c_1 \cos(t) + c_2 \sin(t)) \\ e^{-\delta t}(c_1 \sin(t) - c_2 \cos(t)) \end{pmatrix},$$

for coefficients $c_1, c_2 \in \mathbb{R}$. Moreover, there is an asymptotically stable equilibrium at $x^* = 0$. Now, let $\varepsilon > 0$ be arbitrary. For $x(0) = (0, \varepsilon/2)$ we have $x(0) \in \mathcal{B}(0; \varepsilon)$. Moreover, this choice of $x(0)$ fixes the constants to $c_1 = 0$, $c_2 = -\varepsilon/2$. For $\hat{t} = \pi/2$ we then have $(x_1(\hat{t}), x_2(\hat{t})) = (-2e^{-\delta\pi/4}\varepsilon, 0)$. If $\delta > 0$ is small enough, we have $2e^{-\delta\pi/4} > 1$. In this case, $x(\hat{t}) \notin \mathcal{B}(0; \varepsilon)$. Thus, $\mathcal{B}(0; \varepsilon)$ is not positively invariant.

□

Exercise 2.2 (Stability analysis by determinant and trace).

Consider the linear differential equation $x' = Ax$ for $x \in \mathbb{R}^2$, $A \in \mathbb{R}^{2 \times 2}$. Find the eigenvalues of the matrix A in terms of its trace and determinant. Then indicate the regions in the $\text{tr}(A)$ - $\det(A)$ -plane where the origin is a (stable/unstable) node/saddle/focus/center.

Solution. Computing the zeros of the characteristic polynomial of a 2×2 matrix, we find the eigenvalues of A as $\lambda_{1,2} = \frac{1}{2}\text{tr}(A) \pm \frac{1}{2}\sqrt{(\text{tr}(A))^2 - 4\det(A)}$. Note also, that this gives $\lambda_1 + \lambda_2 = \text{tr}(A)$ and $\lambda_1 \cdot \lambda_2 = \det(A)$.

This allows to find the regions where real/imaginary part of the eigenvalues are positive/zero/negative.

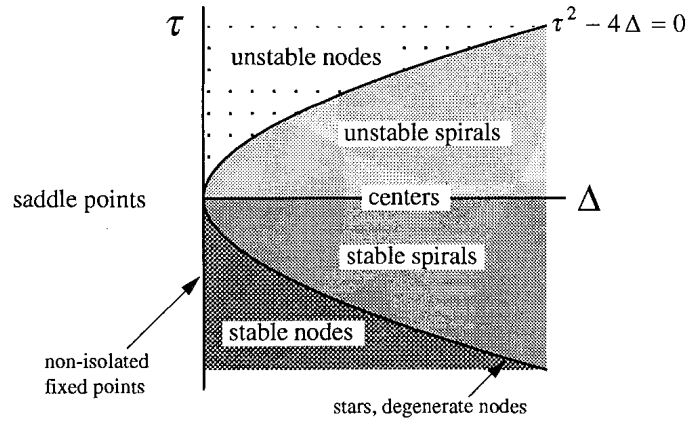


Figure 1: Regions in the $\det(A)$ - $\text{tr}(A)$ -plane where the origin is a (stable/unstable) focus=spiral/node/center/saddle. Here, $\Delta = \det(A)$ and $\tau = \text{tr}(A)$. Taken from *Steven H. Strogatz, Nonlinear Dynamics and Chaos*.

□

Exercise 2.3 (Pendulum).

Consider the (two dimensional) dynamical system induced by the pendulum equation

$$x'' + \sin(x) = 0, \quad x \in \mathbb{R}. \quad (2.1)$$

- a) Show that (2.1) is a Hamiltonian system, i.e. it has the form

$$\begin{aligned} q' &= \partial_p H(q, p) \\ p' &= -\partial_q H(q, p) \end{aligned}$$

for some $H \in C^1(\mathbb{R}^2; \mathbb{R})$. Furthermore, show that for every Hamiltonian system the Hamiltonian H is constant along trajectories, i.e. it is a first integral.

- b) Find a suitable Lyapunov function to show that the equilibrium at $(0, 0)$ is stable.
c) Use the fact that the Hamiltonian H is a first integral to sketch the phase portrait.

Solution.

a) First let us write the second order ODE as a first order system. By setting $y = x'$ we get

$$\begin{aligned}x' &= y, \\y' &= -\sin(x),\end{aligned}$$

so with the Hamiltonian $H(x, y) = \frac{1}{2}y^2 - \cos(x)$ we obtain the required form.

Next, consider some general Hamiltonian system

$$\begin{aligned}q' &= \partial_p H(q, p), \\p' &= -\partial_q H(q, p).\end{aligned}$$

We quickly compute the Lie derivative:

$$\nabla H(q, p) \cdot f(p, q) = \begin{pmatrix} \partial_q H \\ \partial_p H \end{pmatrix} \cdot \begin{pmatrix} \partial_p H \\ -\partial_q H \end{pmatrix} = \partial_q H \partial_p H - \partial_p H \partial_q H = 0$$

which shows that H is a first integral.

b) We take the Lyapunov function $L(x, y) = H(x, y) + 1 = \frac{1}{2}y^2 - \cos(x) + 1$. The function L vanishes at the equilibrium $(0, 0)$ and is strictly positive for all other points in a neighbourhood of $(0, 0)$.

Furthermore, the Lie-derivative

$$\mathcal{L}_f L(x) = \langle \nabla L(x), f(x) \rangle = \sin(x) \cdot y + y \cdot (-\sin(x)) = 0$$

is zero as for every Hamiltonian system, meaning that L is non-increasing along orbits. By Theorem 2.4 the equilibrium point $(0, 0)$ is consequently stable.

c) First note that that the equations (and thus also the phase portrait) is 2π periodic in x -direction and equilibria are located at $(x, y) = (k\pi, 0), k \in \mathbb{Z}$.

As the Hamiltonian is constant along trajectories, the trajectories lie within the level sets of the Hamiltonian $H(x, y) = \frac{1}{2}y^2 - \cos(x)$.

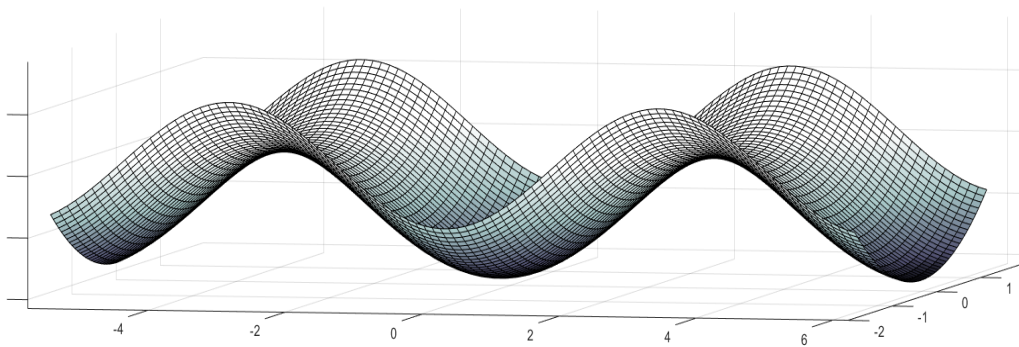


Figure 2: Hamiltonian $H(x, y) = \frac{1}{2}y^2 - \cos(x)$ of the pendulum equation

Since $\dot{x} = y$ the flow is always directed to the right in upper half plane $\{y > 0\}$ and to the left in the lower half plane $\{y < 0\}$.

Equilibria at $(x, y) = (0, 2k\pi), k \in \mathbb{Z}$, correspond to local minima of the Hamiltonian and the level sets lie circle-like around them. On the other hand, equilibria at

$(x, y) = (0, (2k + 1)\pi), k \in \mathbb{Z}$, correspond to saddle points of the Hamiltonian which produce typical saddle point orbits.

This gives the following phase portrait:

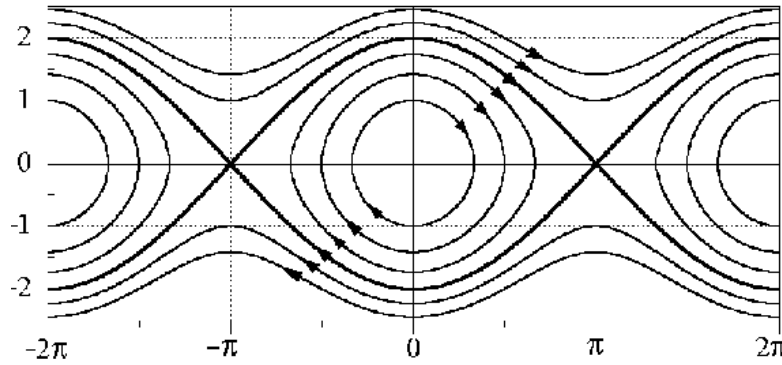


Figure 3: Phase portrait of the pendulum equation, *E. Butikov, The rigid pendulum - An antique but evergreen physical model*

□

Exercise 2.4 (Stable and unstable manifold).

Consider the differential equation

$$\begin{aligned} x' &= x, \\ y' &= -y + x^2. \end{aligned} \tag{2.2}$$

- Show that the origin is a hyperbolic equilibrium point.
- Determine the stable and unstable eigenspaces E^s and E^u of the linearized system.
- Solve (2.2) explicitly to determine the local stable and unstable manifolds $W_{loc}^s(0)$ and $W_{loc}^u(0)$ of (2.2).

Solution.

- Clearly the vector field vanishes at the origin. Further, we compute the derivative of $f(x, y) = \begin{pmatrix} x \\ -y + x^2 \end{pmatrix}$ at the origin as

$$Df(x, y)|_{(x,y)=(0,0)} = \begin{pmatrix} 1 & 0 \\ 2x & -1 \end{pmatrix}|_{(x,y)=(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This matrix has eigenvalues 1 and -1 and since neither of them has zero real part, the origin is a hyperbolic equilibrium point.

- The corresponding eigenvectors to the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$ are $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, so that the unstable eigenspace is given by

$$E^u = \text{span}\{v_1\} = \{(x, y) \in \mathbb{R}^2 \mid y = 0\}$$

and the stable eigenspace is given by

$$E^s = \text{span}\{v_2\} = \{(x, y) \in \mathbb{R}^2 \mid x = 0\}.$$

- c) In our case we can explicitly solve the differential equation. As the x -equation is decoupled we directly get here $x(t) = e^t x_0$ for some initial condition $x(0) = x_0$. Taking this to the second equation we obtain

$$y' = -y + e^{2t} x_0^2.$$

The solution of this linear inhomogeneous equation to the initial condition $y(0) = y_0$ is

$$y(t) = e^{-t} y_0 + \frac{1}{3} e^{2t} x_0^2 - \frac{1}{3} e^{-t} x_0^2.$$

Thus the flow is given by

$$\phi_t(x_0, y_0) = \begin{pmatrix} e^t x_0 \\ e^{-t} y_0 + \frac{1}{3} e^{2t} x_0^2 - \frac{1}{3} e^{-t} x_0^2 \end{pmatrix}.$$

For the stable manifold $W_{loc}^s(0)$ we are looking for those initial conditions (x_0, y_0) for which $\lim_{t \rightarrow \infty} \phi_t(x_0, y_0) = 0$ holds true. As $t \rightarrow \infty$, the first component of $\phi_t(x_0, y_0)$ converges to 0 if and only if $x_0 = 0$. In this case the second component is $e^{-t} y_0$ which converges towards 0 independently of y_0 .

Thus the stable manifold is given by

$$W_{loc}^s(0) = \{(x, y) \in \mathbb{R}^2 \mid x = 0\} = E^s.$$

For the unstable manifold $W_{loc}^u(0)$ on the other hand we need $\lim_{t \rightarrow -\infty} \phi_t(x_0, y_0) = 0$.

The first component of $\phi_t(x_0, y_0)$ converges to 0 regardless of x_0 . For the second component, notice that the term $\frac{1}{3} e^{2t} x_0^2$ tends towards 0 as $t \rightarrow -\infty$ while $e^{-t} y_0$ and $-\frac{1}{3} e^{-t} x_0^2$ diverge. We thus have to choose x_0 and y_0 such that the latter cancel each other, i.e. $y_0 = \frac{1}{3} x_0^2$.

Consequently the unstable manifold is given by

$$W_{loc}^u(0) = \{(x, y) \in \mathbb{R}^2 \mid y = \frac{1}{3} x^2\}.$$

□